

Macro F1 Estimation Reading Synthesis

Research reading artifact | fictional/internal | 2026-07-13

Analytical frame. This synthetic reading synthesis explains the calculation used to contextualize the Helix Forge Lab result on 2026-07-13: Model Alpha M-14, run R184, macro F1 0.873. It is a worked measurement artifact rather than a report of real observations.

Class-level example

Suppose a rare synthetic intent has F1 0.71 while four peers range from 0.84 to 0.91. Macro averaging preserves that warning instead of weighting it away.

Formula derivation

For label k , $F1_k = 2 P_k R_k / (P_k + R_k)$. Macro F1 is the arithmetic mean of the five label scores, not a pooled contingency-table ratio.

Averaging caveats

The mean is sensitive to each class equally. It should be paired with counts and a per-label table whenever a design decision is proposed.

Working matrix

Illustrative label	Precision	Recall	F1
retrieve	0.90	0.86	0.88
route	0.84	0.89	0.86
abstain	0.78	0.65	0.71

Five-label worked calculation

Label	Precision	Recall	F1
retrieve	0.90	0.86	0.88
route	0.84	0.89	0.86
refine	0.87	0.83	0.85
resolve	0.91	0.88	0.89
abstain	0.78	0.65	0.71
Macro average			0.838

Macro versus micro caution panel

Macro F1 gives abstain the same analytical weight as retrieve. A micro-style pooled score can instead be pulled toward the most numerous synthetic label. *Label Arithmetic Notebook* [synthetic/internal] recommends reporting both the formula and the per-label inputs; *Macro-Average Review Primer* [synthetic/internal] treats disagreement between the summaries as a review signal.

Reporting rule

When macro and pooled summaries tell different stories, publish the five-label table first and describe the pooled result as context. This protects the low-F1 abstain example from disappearing into a larger count.